

$N=1$ pure SUGRA:

* SUGRA $\rightarrow$ Local

• local SUSY $\leftrightarrow$ Supergravity

Gravity:

$$\lambda = 2: \left\{ \begin{array}{ccc} 2 & 1 & -2 \\ 3/2 & 1 & -3/2 \end{array} \right\} (N=1)$$
$$(g_{\mu\nu}, \psi_\mu, c_\mu)$$

Noether procedure:

Dirac field $S_0 = i \int d^4x \bar{\psi} \gamma^\mu \partial_\mu \psi$

• $U(1)$ Symmetry:

$$\psi \rightarrow e^{i\varepsilon} \psi \quad | \quad \bar{\psi} = e^{i\varepsilon} \bar{\psi}$$

which is global

• Make local: $\psi \rightarrow e^{i\varepsilon(x)} \psi$

$$\psi' = \psi - i\varepsilon(x) \psi$$

$$\Rightarrow \delta\psi = -i\varepsilon(x) \psi$$

$$\begin{aligned} \delta S_0 &= i \int d^4x \delta\bar{\psi} \gamma^\mu \partial_\mu \psi + i \int d^4x \bar{\psi} \gamma^\mu \partial_\mu \delta\psi \\ &= i \int d^4x (i\varepsilon \bar{\psi}) \gamma^\mu \partial_\mu \psi + i \int d^4x \bar{\psi} \gamma^\mu \partial_\mu (-i\varepsilon \psi) \\ &= - \int d^4x \varepsilon (\bar{\psi} \gamma^\mu \partial_\mu \psi) + \int d^4x \bar{\psi} \gamma^\mu (\psi \partial_\mu \varepsilon + \varepsilon \partial_\mu \psi) \\ &= \int d^4x (\bar{\psi} \gamma^\mu \psi) \partial_\mu \varepsilon \end{aligned}$$

$$\boxed{\delta S_0 = \int d^4x j^\mu \partial_\mu \varepsilon(x)}, \quad \underline{j^\mu = \bar{\psi} \gamma^\mu \psi}$$

* Restore invariance of S_0 : introduce a gauge field A_μ

$$A_\mu \rightarrow A_\mu + \underbrace{\partial_\mu \varepsilon}_{\rightarrow \delta A_\mu}$$

• $\delta j^\mu = 0$

and so

$$\delta S_0 = \int d^4x j^\mu \delta A_\mu$$

$$= \delta \left(\int d^4x j^\mu A_\mu \right) \leftarrow S_A$$

$\leadsto$ must add this exact term to the action

$$S \equiv S_0 - S_A \Rightarrow \delta S = \delta S_0 - \delta S_A = 0 \quad \checkmark$$

$$S = S_0 - \int d^4x j^\mu A_\mu$$

$$* S = \int d^4x \left(i \bar{\psi} \gamma^\mu \partial_\mu \psi - \bar{\psi} \gamma^\mu \psi A_\mu \right)$$

$$S = \int d^4x i \bar{\psi} \gamma^\mu \underbrace{(\partial_\mu + i A_\mu)}_{D_\mu} \psi$$

$$\therefore S = \int d^4x i \bar{\psi} \gamma^\mu D_\mu \psi = i \int d^4x \bar{\psi} \not{D} \psi$$

* Ferm. part of QED

$$A \in \mathcal{A} \otimes \Omega^1(M)$$

$$A_\mu^a : \begin{array}{|c|} \hline \square \\ \hline \end{array} \quad \text{adjoint}$$

$$A = A_\mu^a T^a \otimes dx^\mu$$

$$G_{\mu\nu}^a \equiv \partial_\mu A_\nu^a - \partial_\nu A_\mu^a$$

$$S_g = -\frac{1}{4} \int d^4x G_{\mu\nu}^a G^{\mu\nu a}$$

* Since A_μ^a transforms in $\mathbb{P}$ $[T^a, T^b] = if_{c}^{ab} T^c$

$$(T^a)_{nm} = f_{nm}^a$$

$$\therefore \left(e^{ig \epsilon_\beta T^\beta} A_\mu \right)^a = A_\mu^a - i \left(\epsilon T A_\mu \right)^a$$

$$= A_\mu^a - ig \epsilon_\beta f^{\beta a \gamma} A_\mu^\gamma$$

$$A_\mu^a \rightarrow A_\mu^a + ig f^{\alpha\beta\gamma} \epsilon_\beta A_\mu^\gamma$$

(i) S_{g_0} is invariant under local $U(1)$:
 $A_\mu \rightarrow A_\mu + \partial_\mu \epsilon$

$$\begin{cases} G'_{\mu\nu} = \partial_\mu (A_\nu + \partial_\nu \epsilon) - \partial_\nu (A_\mu + \partial_\mu \epsilon) \\ = \partial_\mu A_\nu + \partial_\mu \partial_\nu \epsilon - \partial_\nu A_\mu - \partial_\nu \partial_\mu \epsilon \\ = G_{\mu\nu} \end{cases}$$

(ii) Not invariant under local non-abel trans.
 but is global

$$\begin{aligned} G'^a_{\mu\nu} &= \partial_\mu (A_\nu^a + ig f^{abc} \epsilon_b A_\nu^c) - \partial_\nu (A_\mu^a + ig f^{abc} \epsilon_b A_\mu^c) \\ &= (\partial_\mu A_\nu^a - \partial_\nu A_\mu^a) + ig f^{abc} \epsilon_b (\partial_\mu A_\nu^c - \partial_\nu A_\mu^c) \\ &= G_{\mu\nu}^a (\delta_c^a + ig f^{abc} \epsilon_b) \end{aligned}$$

$$G'^a_{\mu\nu} G'^{a\mu\nu} = G_{\mu\nu}^a G^{\mu\nu a} (\delta_c^a + ig f^{abc} \epsilon_b) (\delta_a^c + ig f^{dca} \epsilon_d)$$

$$f^{dbc} = -f^{abc}$$

So cross terms vanish, ϵ^2 also zero

$$\therefore G'^a_{\mu\nu} G'^{a\mu\nu}$$

• Make local: $A_\mu^a \rightarrow A_\mu^a + ig f^{abc} \epsilon_b(x) A_\mu^c$

$$G'^a_{\mu\nu} = G_{\mu\nu}^a + ig f^{abc} \partial_\mu \epsilon_b(x) A_\nu^c - ig f^{abc} \partial_\nu \epsilon_b(x) A_\mu^c$$

$$\begin{aligned} G'^a_{\mu\nu} G'^{a\mu\nu} &= (G_{\mu\nu}^a + \Sigma_{\mu\nu}^a) (G^{\mu\nu a} + \Sigma^{\mu\nu a}) \\ &= GG + 2 \Sigma_{\mu\nu}^a G^{\mu\nu a} \end{aligned}$$

$$G'^a_{\mu\nu} = G_{\mu\nu}^a (\delta_c^a + ig f^{abc} \epsilon_b) + \text{terms with } \partial \epsilon$$

$\rightarrow$

$$\begin{aligned} + f^{abc} \partial_\nu \epsilon_b A_\mu^c &= -f^{acb} \partial_\nu \epsilon_c A_\mu^b \\ &= f^{abc} \partial_\nu \epsilon_c A_\mu^b \end{aligned}$$

$$= if^{abc} g (\partial_\mu \epsilon^b A_\nu^c + \partial_\nu \epsilon^c A_\mu^b)$$

$$G'^{\mu\nu a} = G^{\mu\nu a} (\delta + ig) + if^{abc} g (\quad)$$

$$G^{\mu\nu a} if^{abc} g (\quad) + G_a^{\mu\nu} if^{abc} (\quad)$$

$$\begin{aligned} & G_a^{\mu\nu} f^{abc} g (\partial_\mu \xi_b A_\nu^c - \partial_\nu \xi_b A_\mu^c) \\ &= 2 f^{abc} g G_a^{\mu\nu} \partial_\mu \xi_b A_\nu^c \\ &= 4 f^{abc} g G_a^{\mu\nu} \partial_\mu \xi_b A_\nu^c \end{aligned}$$

$$= (-\frac{1}{4}) (-4 f^{bac} G_a^{\mu\nu} A_\nu^c) \partial_\mu \xi_b$$

$ac \rightarrow b$

$$= \int d^4x (g f^{abc} G_b^{\mu\nu} A_\nu^c) \partial_\mu \xi_a(x)$$

$$\int d^4x j^{\mu a} \partial_\mu \xi_a$$

$$j^{\mu a} = g f^{abc} G_b^{\mu\nu} A_\nu^c$$

$$S_{g,1} = S_{g,0} - \frac{1}{2} \int d^4x j^{\mu a} A_{\mu a}$$

$$A_a^\mu \rightarrow A_a^\mu + g f^{abc} \xi_b A_c^\mu + \partial^\mu \xi_a$$

$g \quad U(1)$

$$G \cong g \times U(1)$$

- $S_{g,1}$ invariant up to order g :
- $\mathcal{I}_{UV}$ up to order g^2 requires

$$\delta S_{1,g} = -g^2 f^{abc} f^{bac} \int A_a^\mu A_\nu^a A_c^\nu \partial^\mu \xi_b d^4x$$

$$\begin{aligned} \delta S_{1,g} &= \delta S_{g,0} - \frac{1}{2} \int d^4x \delta(j^{\mu a} A_{\mu a}) \\ &= \int d^4x j^{\mu a} \partial_\mu \xi_a - \frac{1}{2} \int d^4x j^{\mu a} (g f^{abc} \xi_b A_c^\mu + \partial^\mu \xi_a) \end{aligned}$$

g^2 term

Assuming $\delta j^{\mu\nu} = 0$

$$q^2: -\frac{1}{2} \int d^4x (g f^{abc} G_b^{\mu\nu} A_\nu^c) (g f_{ade} \xi_d A_\mu^e)$$

$$= -\frac{1}{2} g^2 f^{abc} f_{ade} \int d^4x (\partial^\mu A_b^\nu - \partial^\nu A_b^\mu) \xi_d A_\mu^e A_\nu^c$$

$\leadsto$ Compact Supp $\Rightarrow$ Boundary terms $\xrightarrow{x \rightarrow \infty} 0$

$$= -\frac{1}{2} g^2 f^{abc} f_{ade} \int d^4x \left(-\partial^\mu \xi_d A_b^\nu A_\mu^e A_\nu^c + \partial^\nu \xi_d A_b^\mu A_\mu^e A_\nu^c \right)$$

$$= -\frac{1}{2} g^2 f^{abc} f_{ade} \int \partial^\mu \xi_d \left(-A_b^\nu A_\mu^e A_\nu^c + A_b^\nu A_\nu^e A_\mu^c \right)$$

$$= -\frac{1}{2} g^2 f^{abc} f_{ade} \int \partial^\mu \xi_d A_b^\nu (A_\nu^e A_\mu^c - A_\mu^e A_\nu^c)$$

$$= \frac{1}{2} g^2 f^{abc} f_{bde} \int \partial^\mu \xi_d A_a^\nu (A_\nu^e A_\mu^c - A_\mu^e A_\nu^c)$$

$$a \leftrightarrow c$$

$$abc = cab = -cba$$

$$= -\frac{1}{2} g^2 f^{cba} f_{bde} \int \partial^\mu \xi_d A_a^\nu (A_\nu^e A_\mu^c - A_\mu^e A_\nu^c)$$

$$a \leftrightarrow c$$

$$= \frac{1}{2} g^2 f^{abc} f_{bde} \int \partial^\mu \xi_d A_c^\nu (A_\nu^e A_\mu^a - A_\mu^e A_\nu^a)$$

Second term: $e \leftrightarrow a$: get (1) -

$$= -g^2 f_{abc} f_{bde} \int d^4x A_\mu^a A_\nu^e A_c^\nu \partial^\mu \xi_d$$

as claimed.

$$\frac{1}{4} \delta S_{g,1} \rightarrow \text{Add to } L \quad \partial^\mu \xi_d \rightarrow A_d^\mu$$

$$S_{Y.M} = S_{g,1} + \frac{1}{4} g^2 f_{abc} f_{bde} \int d^4x A_\mu^a A_\nu^e A_c^\nu A_d^\mu$$

• Non-Ab. P.M.

$$\psi \rightarrow e^{i\varepsilon} \psi = \psi - i\varepsilon \psi$$

$$[*] \int d^4x j^\mu \partial_\mu \varepsilon, \quad (-\partial_\mu \varepsilon) \rightarrow A_\mu$$

$$* S = -\frac{1}{4} \int d^4x G^{\mu\nu} G_{\mu\nu} - \frac{1}{2} \int d^4x j^\mu A_\mu + \frac{1}{4} ()$$

$$S = -\frac{1}{4} \int d^4x F_a^{\mu\nu} F_{\mu\nu}^a$$

$$* F_c^{\mu\nu} = \partial^\mu A_c^\nu - \partial^\nu A_c^\mu - g [A \wedge A]_c^{\mu\nu}$$

$$F_c^{\mu\nu} = \partial^\mu A_c^\nu - \partial^\nu A_c^\mu - g A_a^\mu A_b^\nu f^{abc}$$

(Non-Ab. Gauge theory)

* Globally super sym & of supergravity

• Supergravity multiplet (D=4, N=1)
($\psi_\mu^\alpha, g_{\mu\nu}, \lambda_\mu$) (off-shell)

$$(\psi_\mu^\alpha, g_{\mu\nu}) \leftarrow (\text{on-shell})$$

$$* \partial^2 h_{\mu\nu} = 0 = \partial^\mu h_{\mu\nu} \quad \left(\begin{array}{l} \text{linearized} \\ \text{gravity} \end{array} \right)$$

and $\eta^{\mu\nu} h_{\mu\nu} = 0$

$$\not{\partial} \psi = 0 = \gamma^\mu \psi_\mu = \partial^\mu \psi_\mu$$

* Massless (free):

$$\text{Spin } 0 \quad \partial^2 \phi = 0$$

$$\text{Spin } 1/2 \quad \not{\partial} \lambda = 0$$

$$\text{Spin } 1 \quad \partial^\mu A_\mu = 0 \quad \leftarrow \text{Lorentz Gauge } \partial^\mu F_{\mu\nu} = 0$$

$$\text{Spin } 3/2 \quad \not{\partial} \psi = 0$$

$$\text{Spin } 2 \quad \partial^2 h_{\mu\nu} = 0$$

→ Can determine general form of E.O.M. under their rep of $ISO^+ \times \mathbb{R}^4$ and invariance:

$$\bullet R^\mu = \epsilon^{\mu\nu\rho\kappa} i \gamma_5 \gamma_\nu \partial_\rho \psi_\kappa \quad \Bigg| \quad \psi \rightarrow i\psi$$

$$\bullet \{ \gamma^\mu, \gamma^\nu \} = 2\eta^{\mu\nu} \mathbb{1}$$

$$\gamma_5 = -i \gamma^0 \gamma^1 \gamma^2 \gamma^3 \quad \text{West}$$

$$\epsilon^{0123} = -1$$

$$S_{RS} = -\frac{i}{2} \int d^4x \epsilon^{\mu\nu\rho\sigma} \bar{\psi}_\mu \gamma_5 \gamma_\nu \partial_\rho \psi_\sigma$$

* E.H. action for gravity must be quadratic in h to apply to our massless limit

$$g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu} \quad \kappa^2 = 8\pi G_N$$

Proof: Use $\delta R_{\mu\nu} = \nabla_\rho \delta \Gamma^\rho_{\mu\nu} - \nabla_\mu \delta \Gamma^\rho_{\nu\rho}$

$$S_{E.H.} = \frac{1}{\kappa^2} \int d^4x \sqrt{|g|} R$$

$$R = g^{\mu\nu} R_{\mu\nu}$$

$$\bullet g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$$

$$R_{\mu\nu} = R_{\mu\sigma\nu}{}^\sigma$$

$$R_{\mu\sigma\nu}{}^\rho = \partial_\mu \Gamma^\rho_{\sigma\nu} - \partial_\sigma \Gamma^\rho_{\mu\nu} + \Gamma^\rho_{\mu\tau} \Gamma^\tau_{\sigma\nu} - \Gamma^\rho_{\sigma\tau} \Gamma^\tau_{\mu\nu}$$

$$\bullet R_{\mu\sigma\nu}{}^\rho = R^\rho{}_{\nu\mu\sigma} = -R_{\nu}{}^\rho{}_{\mu\sigma}$$

$$\rho \leftrightarrow \sigma \quad \mu \leftrightarrow \nu$$

$$= -R_{\mu}{}^\rho{}_{\nu\rho}$$

$$\Gamma^\rho_{\mu\tau} = \frac{1}{2} g^{\rho\alpha} (\partial_\mu g_{\tau\alpha} + \partial_\tau g_{\mu\alpha} - \partial_\alpha g_{\mu\tau})$$

$$g_{\mu\nu} g^{\mu\nu} = (\eta_{\mu\nu} + \kappa h_{\mu\nu})(\eta^{\mu\nu} - \kappa h^{\mu\nu}) = 4$$

$$= \eta^2 + \kappa h - \kappa h - \kappa^2 h^2$$

$$h^2 = 0$$

$$\begin{aligned} \Gamma_{\mu\tau}^{\rho} &= \frac{\kappa}{2} (\eta^{\rho\alpha} - \kappa h^{\rho\alpha}) (\partial_{\mu} h_{\alpha\tau} + \partial_{\tau} h_{\alpha\mu} - \partial_{\alpha} h_{\mu\tau}) \\ &= \frac{\kappa}{2} \eta^{\rho\alpha} (\partial_{\mu} h_{\alpha\tau} + \partial_{\tau} h_{\alpha\mu} - \partial_{\alpha} h_{\mu\tau}) \end{aligned}$$

$$R_{\mu\sigma}^{\quad L \rho \quad \nu} = \partial_{\mu} \Gamma_{\sigma\nu}^{\rho} - \partial_{\sigma} \Gamma_{\mu\nu}^{\rho}$$

$$\begin{aligned} R_{\mu\sigma}^{\quad L \rho \quad \nu} &= \frac{\kappa}{2} \eta^{\rho\alpha} (\partial_{\mu} \partial_{\sigma} h_{\alpha\nu} + \partial_{\mu} \partial_{\nu} h_{\sigma\alpha} - \partial_{\mu} \partial_{\alpha} h_{\sigma\nu} \\ &\quad - \partial_{\sigma} \partial_{\mu} h_{\alpha\nu} - \partial_{\sigma} \partial_{\nu} h_{\mu\alpha} + \partial_{\sigma} \partial_{\alpha} h_{\mu\nu}) \end{aligned}$$

$$R_{\mu\sigma}^{\quad L \rho \quad \nu} = \frac{\kappa}{2} \eta^{\rho\alpha} (\partial_{\mu} \partial_{\nu} h_{\sigma\alpha} + \partial_{\sigma} \partial_{\alpha} h_{\mu\nu} - \partial_{\sigma} \partial_{\nu} h_{\mu\alpha} - \partial_{\mu} \partial_{\alpha} h_{\sigma\nu})$$

$$R_{\mu\nu}^{\quad L} = -R_{\mu\sigma}^{\quad L \sigma \quad \nu}, \quad h \equiv \eta^{\sigma\lambda} h_{\sigma\lambda}$$

$$= \frac{\kappa}{2} (-\partial_{\mu} \partial_{\nu} h - \partial_{\sigma} \partial^{\sigma} h_{\mu\nu} + \partial^{\rho} \partial_{\nu} h_{\mu\rho} + \partial_{\mu} \partial^{\rho} h_{\rho\nu})$$

$$R_{\mu\nu}^{\quad L} = \frac{\kappa}{2} (\partial_{\mu} \partial^{\rho} h_{\nu\rho} + \partial_{\nu} \partial^{\rho} h_{\mu\rho} - \partial^2 h_{\mu\nu} - \partial_{\mu} \partial_{\nu} h)$$

$$R^L \equiv \eta^{\mu\nu} R_{\mu\nu}^{\quad L}$$

$$\begin{aligned} \det(g_{\mu\nu}) &= \epsilon_{\mu_1 \dots \mu_4} g^{\mu_1 1} \dots g^{\mu_4 4} \\ &= \epsilon_{\mu_1 \dots \mu_4} (\eta + \kappa)^1 \dots (\eta + \kappa)^4 \\ &= (\eta + \kappa)^4 = (\eta + 2\eta\kappa)(\eta + \kappa) \\ &= 3\eta\kappa \end{aligned}$$

$$\begin{aligned} &4\eta\kappa h + \eta^2 \\ &= 4(1 + \kappa\eta^{\mu\nu} h_{\mu\nu}) \end{aligned}$$

$$\sqrt{\det g} = 2(1 + \kappa \eta^{\mu\nu} h_{\mu\nu})$$

• Not quite right

$$\sqrt{|g|} = (1 + \frac{1}{2} \kappa h^\mu_\mu + \mathcal{O}(h^2)) \sqrt{|h|}$$

$$\frac{1}{2\kappa^2} \int d^4x \sqrt{|g|} R \rightarrow \frac{1}{2} \int d^4x (1 + \frac{\kappa}{2} h^\mu_\mu) (\eta^{\mu\nu} - \kappa h^{\mu\nu}) R^\mu_\nu$$

$$\frac{1}{2} \int d^4x (\eta^{\mu\nu} + \frac{\kappa}{2} h^\mu_\sigma \eta^{\sigma\nu} - \kappa h^{\mu\nu}) R^\mu_\nu$$

$$\kappa^2: \frac{1}{2} \int d^4x (R^\mu_\mu + \frac{\kappa}{2} h^\mu_\sigma R^\sigma_\mu - \kappa h^{\mu\nu} R^\mu_\nu)$$

$$S_E^L = -\frac{1}{2} \int d^4x (R^\mu_\mu - \frac{1}{2} \eta_{\mu\nu} R^\mu_\mu) h^{\mu\nu}$$

* On-shell, global susy action

$$S_L = S_{RS} + S_E^L$$

→ Susy closes on E.O.M

$$\underline{R^\mu_\mu = 0}, \quad \underline{\epsilon^{\mu\nu\rho\sigma} \gamma_5 \gamma_\nu \partial_\rho \psi_\sigma = 0}$$

$$\begin{array}{l} * \delta_\eta S_{RS} = 0 \\ \quad \underline{\delta_\eta \psi_\mu = \partial_\mu \eta} \end{array} \quad \Bigg| \quad \eta: \text{Majorana}$$

$$* \delta_\xi S_E^L = 0, \quad \text{local diffeomorphism}$$

$$\underline{\delta_\xi h_{\mu\nu} = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu}$$

$$\rightarrow X'^\mu = X^\mu + \xi^\mu(x)$$

* Now, consider a Global Susy $\mathcal{E}^{1|5Q}$

• Lorentz symm $\oplus$ dim analysis

$$h_{\mu\nu} \rightarrow h_{\mu\nu} + c_1 \bar{\xi} (\gamma_\mu \psi_\nu + \gamma_\nu \psi_\mu) + c_2 \eta_{\mu\nu} \bar{\xi} \gamma^p \psi_p$$

$$= h_{\mu\nu} + \delta_{\bar{\xi}} h_{\mu\nu}$$

$$\psi_\mu \rightarrow \psi_\mu + c_3 \epsilon^{\rho\tau} \partial_\rho h_{\tau\mu} \bar{\xi} + c_4 \partial_\rho h_{\mu}^{\rho} \bar{\xi} = \psi_\mu + \delta_{\bar{\xi}} \psi_\mu$$

$$\sigma^{\mu\nu} = \frac{i}{2} [\gamma^\mu, \gamma^\nu]$$

* Determine these constants by assuming these all form a closed algebra

(i) $[\delta_\eta, \delta_{\bar{\xi}}] h_{\mu\nu}$

$$(\delta_\eta \delta_{\bar{\xi}} - \delta_{\bar{\xi}} \delta_\eta) h_{\mu\nu} = \delta_\eta (\delta_{\bar{\xi}} h_{\mu\nu})$$

$$= \delta_\eta (c_1 \bar{\xi} (\gamma_\mu \psi_\nu + \gamma_\nu \psi_\mu) + c_2 \eta_{\mu\nu} \bar{\xi} \gamma^p \psi_p)$$

$$= c_1 \bar{\xi} (\gamma_\mu \delta_\eta \psi_\nu + \gamma_\nu \delta_\eta \psi_\mu) + c_2 \eta_{\mu\nu} \bar{\xi} \gamma^p \delta_\eta \psi_p$$

$$= c_1 \bar{\xi} (\gamma_\mu \partial_\nu \eta + \gamma_\nu \partial_\mu \eta) + c_2 \eta_{\mu\nu} \bar{\xi} \gamma^p \partial_p \eta$$

• Take c_2 to be zero so that this becomes a diffeomorphism:

(Q: Could we not take a conformal spacetime to have the extra factor?

$$\bullet \partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu - \frac{2}{D} \eta_{\mu\nu} \partial_\rho \epsilon^\rho$$

Identify: $\epsilon_\mu = c_1 \bar{\xi} \gamma_\mu \eta$

Then: $c_2 \eta_{\mu\nu} \bar{\xi} \gamma_p \partial^p \eta = c_2 \partial^p (\bar{\xi} \gamma_p \eta) \eta_{\mu\nu}$
 $= (c_2/c_1) \eta_{\mu\nu} \partial^p \epsilon_p$

$$\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu + \frac{c_2}{c_1} \eta_{\mu\nu} \partial^p \epsilon_p$$

• how would $h_{\mu\nu}$ look? can the conf. factor be a

$$h_{\mu\nu} = \Omega^2(x) \eta_{\mu\nu}$$

$$R'^{\mu\nu} = \frac{1}{2} \left(-\partial^2 (e^\phi h_{\mu\nu}) + \partial_\mu \partial_\lambda h^\lambda{}_\nu e^\phi + \partial_\nu \partial_\lambda e^\phi h^\lambda{}_\mu - \partial_\mu \partial_\nu (e^\phi h) \right)$$

$$(i) R'^{\mu\nu} = R_{\mu\nu} - 2 (\partial_\mu \partial_\nu \phi - \partial_\mu \phi \partial_\nu \phi) = (\partial^2 \phi + 2(\partial\phi)^2) \eta_{\mu\nu}$$

$$\eta^{\mu\nu} R'_{\mu\nu} = R^2 - 2 (\partial^2 \phi + 2(\partial\phi)^2) = R^2 - 6\partial^2 \phi + 6(\partial\phi)^2$$

$$R^2 = R'^2 + 6\partial^2 \phi - 6(\partial\phi)^2$$

$\Rightarrow$ this is not invariant under conf.

so this theory does not have a conf. symm
 $\neq S_0$, that's why we can't let $c_2 \neq 0$.

$$\therefore, \quad c_2 = 0$$

$$[\delta_\eta, \delta_\xi] h_{\mu\nu} = \partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu$$

$$\epsilon_\mu = \bar{\xi} \gamma_\mu \mathcal{N}$$

$$[\delta_\epsilon, \delta_\xi] \psi_\mu = (\delta_\epsilon \delta_\xi - \delta_\xi \delta_\epsilon) \psi_\mu$$

$$\delta_\epsilon \psi_\mu = 0$$

$$= \delta_\epsilon (c_3 \sigma^{\rho\tau} \partial_\rho h_{\tau\mu} \xi + c_4 \partial_\rho h^\rho{}_\mu \xi)$$

$$= c_3 \sigma^{\rho\tau} \partial_\rho (\delta_\epsilon h_{\tau\mu}) \xi + c_4 \partial_\rho (\delta_\epsilon h^\rho{}_\mu) \xi$$

$$= c_3 \sigma^{\rho\tau} \partial_\rho [\partial_\tau \epsilon_\mu + \partial_\mu \epsilon_\tau] \xi + c_4 \eta^{\lambda\rho} \partial_\rho [\partial_\lambda \epsilon_\mu + \partial_\mu \epsilon_\lambda] \xi$$

$$= c_3 \sigma^{\rho\tau} \partial_\rho \partial_\mu \epsilon_\tau \xi + c_4 (\partial^2 \epsilon_\mu + \partial_\mu \epsilon^\mu) \xi$$

$$c_4 = 0$$

$$= \partial_\mu (c_3 \sigma^{\rho\tau} \partial_\rho \epsilon_\tau \xi)$$

$$\mathcal{N} \equiv c_3 \sigma^{\rho\tau} \partial_\rho \epsilon_\tau \xi$$

• SUSY closes on / to spacetime translations:

$$\rightarrow \{Q_\alpha, \bar{Q}_{\dot{\alpha}}\} = i \sigma^\mu_{\alpha\dot{\alpha}} P_\mu$$

so

$$\delta_\xi \psi = (i \xi Q) \psi$$

$$\Rightarrow [\delta_{\xi_1}, \delta_{\xi_2}] = i \bar{\xi}_1 \xi_2 \{Q_\alpha, \bar{Q}_\alpha\}$$

So we should expect this to be a spacetime translation

Two sys:

$$[\delta_{\xi_1}, \delta_{\xi_2}] h_{\mu\nu}$$

$$\delta_{\xi_1}(\delta_{\xi_2} h_{\mu\nu}) = \delta_{\xi_1} (c_2 \bar{\xi}_2 (\gamma_\mu \psi_\nu + \gamma_\nu \psi_\mu))$$

$\delta_{\xi_2} h_{\mu\nu}$ looks like a 1.5 fermion with fermionic version

$$= c_2 \bar{\xi}_2 (\gamma_\mu \delta_{\xi_1} \psi_\nu + \gamma_\nu \delta_{\xi_1} \psi_\mu)$$

$$= c_2 \bar{\xi}_2 [\gamma_\mu c_3 \sigma^{\rho\tau} \partial_\rho h_{\tau\nu} \xi_1 + \gamma_\nu c_3 \sigma^{\rho\tau} \partial_\rho h_{\tau\mu} \xi_1]$$

$$- c_2 \bar{\xi}_1 [c_3 \sigma^{\rho\tau} (\partial_\rho h_{\tau\nu} \gamma_\mu + \gamma_\nu \partial_\rho h_{\tau\mu}) \xi_2]$$

$$= c_2 c_3 \sigma^{\rho\tau} [\bar{\xi}_2 \gamma_\mu \partial_\rho h_{\tau\nu} \xi_1 - \bar{\xi}_1 \partial_\rho h_{\tau\nu} \gamma_\mu \xi_2] + (\mu \leftrightarrow \nu)$$

$$\lambda = \lambda^T C$$

$$\bar{\lambda} \gamma_{\mu_1 \dots \mu_r} \chi = \text{tr} \bar{\chi} \gamma_{\mu_1 \dots \mu_r} \lambda$$

C : Unitary

$$C^T C = \mathbb{1}$$

Proof:

$$(C \gamma^{(r)})^T = -\text{tr} C \gamma^{(r)}$$

$r=0$:

$$C^T = -t_0 C$$

$$= -C$$

$r=1$:

$$(C \gamma^\mu)^T = -t_1 C \gamma^\mu$$

$$= (\gamma^\mu)^T C^T = -t_1 C \gamma^\mu$$

$$= (\gamma^\mu)^T (-t_0 C) = -t_1 C \gamma^\mu$$

$$\Rightarrow (\gamma^\mu)^T = t_0 t_1 C \gamma^\mu C^{-1} = -C \gamma^\mu C^{-1}$$

$$\bar{\xi}_2 \gamma_\mu \xi_1 = \xi_2^T C \gamma_\mu \xi_1$$

$$= \xi_2^T (C \gamma^\mu C^\dagger) (C \xi_1)$$

$$= \xi_2^T [- (\gamma^\mu)^T] (C \xi_1)$$

$$= - \xi_2^T (\gamma^\mu)^T$$

$$[(\xi_1^T C)^T] = C \xi_1$$

$$= \xi_2^T (+ \gamma^\mu)^T (C \xi_1)^T$$

$$= - (\xi_1^T C \gamma^\mu \xi_2)^T$$

$$= - (\bar{\xi}_1 \gamma^\mu \xi_2)^T$$

• Claim: This thing must be invariant under T

$$\bar{\xi}_2 \gamma^\mu \xi_1 = - (\bar{\xi}_1 \gamma^\mu \xi_2)^T$$

$\Rightarrow$

$$= C_1 C_3 \text{ OPE } (- (\bar{\xi}_1 \gamma^\mu \xi_2)^T) \text{ OPE } + \xi_1$$

* lets review formal spinors: (Even d.m)

Spin(1, D-1): Group generated by $\frac{i}{2} \gamma_{mn}$

$$e^{\frac{i}{4} \omega^{mn} \gamma_{mn}}$$

$$\in \text{Spin}(1, D-1)$$

$$\Rightarrow \simeq \mathbb{1} + \frac{i}{4} \omega^{mn} \gamma_{mn}, \quad \omega^{mn} = -\omega^{nm}$$

• Assume Spin(1, D-1) connected to $\mathbb{1}$

i.e. $\exists t_0$ s.t.

$$A(t_0) = \mathbb{1}$$

$$\bullet \det(A) = e^{\text{tr} \ln(A)} = e^{\frac{i}{4} \omega^{mn} \text{tr}(\gamma_{mn})}$$

$$\text{tr}(\gamma_{mn}) = 0$$

$$\text{proof: } \text{Tr}(\gamma_{mn}) = \frac{1}{2} (\text{Tr}(\gamma_m \gamma_n) - \text{Tr}(\gamma_n \gamma_m))$$

$$= \frac{1}{2} (\text{Tr}(\gamma_m \gamma_n) - \text{Tr}(\gamma_m \gamma_n))$$

$$= 0$$

$\therefore$

$$\det(A) = 1$$

Let $V \in C_0$, $V = V^m \gamma_m$

$V \mapsto V' = A V A^{-1}$

is preserved for any $A \in \text{Spin}(1, D-1)$

(i) $[\frac{1}{2} \gamma_{mn}, \gamma_p] = \eta_{pn} \gamma_m - \eta_{pm} \gamma_n$

$A = \mathbb{1} + \frac{1}{4} \omega^{mn} \gamma_{mn}$

$$\begin{aligned} V' &= (\mathbb{1} + \frac{1}{4} \omega^{mn} \gamma_{mn}) V^p \gamma_p (\mathbb{1} - \frac{1}{4} \omega^{qr} \gamma_{qr}) \\ &= (V^p \gamma_p + \frac{1}{4} \omega^{mn} V^p \gamma_{mn} \gamma_p) (\mathbb{1} - \frac{1}{4} \omega^{qr} \gamma_{qr}) \\ &= (V^p \gamma_p + \frac{1}{4} \omega^{mn} V^p \gamma_{mn} \gamma_p - \frac{1}{4} V^p \gamma_p \omega^{qr} \gamma_{qr} + \mathcal{O}(\omega^2)) \\ &= V^p \gamma_p + \frac{1}{2} \omega^{mn} V^p (\gamma_{mn}, \gamma_p) \\ &= V^p \gamma_p + \frac{1}{2} \omega^{mn} V^p (\eta_{pn} \gamma_m - \eta_{pm} \gamma_n) \\ &= V^p \gamma_p + \frac{1}{2} \omega^{mn} V_n \gamma_m - \frac{1}{2} \omega^{mn} V_m \gamma_n \\ &= V^p \gamma_p + \omega^{pn} V_n \gamma_p \end{aligned}$$

$\therefore (V')^p = V^p + \omega^{pn} V_n$ (Inf. Lorentz)

$e^B e^C e^{-B} = \mathbb{1} + [B, C] + \frac{1}{2} [B, [B, C]]$

$A V A^{-1} = \mathbb{1} + [\frac{1}{4} \omega^{mn} \gamma_{mn}, V] + \frac{1}{2} [\frac{1}{4} \omega^{mn} \gamma_{mn}, ()]$
 $= \mathbb{1} + \frac{1}{4} \omega^{mn} V (\gamma_{mn}) + \frac{1}{2} \cdot$

all extra terms add a γ term

so $A V A^{-1}$ preserves the form of $V \forall A$.

$\Rightarrow \text{Spin}(1, D-1) \cong \left\{ \sum_{n_1 \dots n_p} c_{n_1 \dots n_p} \gamma^{n_1 \dots n_p} \mid \text{s.t. } A V A^{-1} \text{ is preserved} \right\}$

Thm: $\text{Spin}(1, D-1) / \mathbb{Z}_2 \cong \text{ISO}(1, D-1)^+$

Proof: Let $A \in \text{Spin}(1, D-1)$

$\rightarrow$ Then we have a map on $\mathbb{R}^{(1, D-1)}$

$A V A^{-1} = \Lambda V$

• Coordinates $V^m = \Lambda^m_n V^n$

→ We've already had a "map" by definition:

$$V^m \partial_m \in \mathbb{R}^{(1, D-1)} \text{ and took}$$

$$\phi(\partial_m) := \gamma_m$$

$$\phi: \mathbb{R}^{(1, D-1)} \rightarrow \text{Span}(1, D-1)$$

• Lorentz invariance

$$V'^2 = V^2 \Leftrightarrow \Lambda \eta \Lambda^T = \eta$$

$$\rightarrow \phi(V'^2) = \phi(V^2)$$

→ must know how ϕ acts with
 $g(V^m \partial_m, V^n \partial_n)$

→ $\phi^* g$: or simply

$$V'^m V'_m = \Lambda^m_n V^n \Lambda^p_m V_p$$

$$2 \gamma^\mu \gamma_\mu = 2 D \mathbb{1}$$

$$\Rightarrow \text{Tr}(\gamma^\mu \gamma_\mu) = D \text{Tr}(\mathbb{1})$$

$$= D \cdot 2^{D/2}$$

$$= D 2^{D/2}$$

$$2^{D/2} = \dim$$

$$\therefore V'^m V'_m = V^m V_m$$

Hence Λ is a Lorentz transformation

$$\phi: \text{Span}(1, D-1) \rightarrow \text{FSO}(1, D-1)^\dagger$$

$$A \mapsto \Lambda_A, \quad V_1 = A_1 V A_1^{-1} \text{ and } V_2 = A_2 V A_2^{-1}$$

$$V_{12} \xrightarrow{A_1} V_1 \xrightarrow{A_2} V_{12}$$

$$V_{12} = A_2 A_1 V A_1^{-1} A_2^{-1} = A_2 A_1 V (A_2 A_1)^{-1}$$

$$= A_2 V_1 A_2^{-1}$$

$$V^m = \Lambda^m_n V^n$$

$$A_1 A_2 \rightarrow \Delta_{A_1} \Delta_{A_2}$$

Thus the map is a homomorphism.

→ Find Kernel for $\neq \mathbb{I}$ and $\rightarrow \mathbb{Z}_2$

Dirac Conjugate: $\bar{\lambda}^D \lambda$ is invariant under $\text{Span}(\mathbb{Z}, D-1)/\mathbb{Z}_2$

$$\delta \bar{\lambda}^D = \bar{\lambda}^D \left[-\frac{i}{4} \omega^{mn} \gamma_{mn} \right]$$

$$\bar{\lambda}^D = \lambda^\dagger \gamma^0$$

Majorana Conjugate $\bar{\lambda}^M$

Defined as $\bar{\lambda}^M \equiv \lambda^T C$

$$\gamma_{mn}^T = -C \gamma_{mn} C^{-1}$$

$$\delta \bar{\lambda}^M = \delta \lambda^T C = (\delta \lambda)^T C$$

$$= \left(\frac{i}{4} \omega^{mn} \gamma_{mn} \lambda \right)^T C$$

$$= \frac{i}{4} \omega^{mn} \lambda^T (\gamma_{mn})^T C$$

$$= -\frac{i}{4} \omega^{mn} \lambda^T C \gamma_{mn} C^{-1} C$$

$$= \bar{\lambda}^M \left(-\frac{i}{4} \omega^{mn} \gamma_{mn} \right)$$

• Maj. changes just as a Dirac spinor of $\text{Span}(\mathbb{Z}, D-1)$

$$\begin{aligned} (\gamma^0)^2 &= -\mathbb{I} \\ C^T &= -C \end{aligned}$$

$$\bar{\lambda}^D = \bar{\lambda}^M$$

$$\lambda^\dagger \gamma^0 = \lambda^T C$$

$$\Rightarrow (\gamma^0)^T \lambda^* = C^T \lambda$$

$$-C \gamma^0 C^{-1} \lambda^* = -C \lambda$$

$$\gamma^0 C^{-1} \lambda^* = \lambda$$

$$C^{-1} \lambda^* = -\gamma^0 \lambda \Rightarrow$$

$$\lambda^* = -C \gamma^0 \lambda$$

$$* C = B^T \gamma^0$$

$$\Rightarrow C^T = -C \Rightarrow (\gamma^0)^T B = -B^T \gamma^0$$

$$\bullet (\gamma^0)^T (\gamma^0)^T = -\mathbb{I}$$

$$-B = -C \gamma^0 C^{-1} B^T \gamma^0$$

$$\mathbb{I}$$

$$-B = -(B^T \gamma^0) \gamma^0 \Rightarrow$$

$$B = -B^T$$

$$\lambda^* = -B^T \gamma^0 \gamma^0 \lambda \rightarrow \boxed{\lambda^* = B \lambda}$$

• $\lambda = B^{-1} \lambda^*$ transforms under $\text{Spin}(1, D-1)$

$$\lambda = B^* \lambda^* = B^* B \lambda = \epsilon \lambda$$

• Major. can only exist if $\epsilon = \pm 1$

$$\Rightarrow D = 2, 4 \bmod (8)$$

$$D = 2, 4, 10, 12, \dots$$

$$* \lambda_\alpha, \quad \alpha = 1, 2, \dots, 2^{D/2}$$

$$\bar{\lambda}^\alpha, \quad \alpha = 1, 2, \dots, 2^{D/2}$$

$$\Rightarrow \bar{\lambda}^\alpha (\gamma_\mu)^\beta_\alpha \lambda_\beta$$

$$\bar{\lambda}^\alpha = (\lambda^T C)^\alpha = \lambda_\beta^T C^{\beta\alpha}$$

$$C^{\alpha\beta} \Leftrightarrow (C^{-1})_{\alpha\beta}$$

$$\bar{\chi} \gamma_{m_1 \dots m_p} \lambda = \chi_\alpha^T (C \gamma_{m_1 \dots m_p})^{\alpha\beta} \lambda_\beta$$

$$(C \gamma^{(r)})^T = -\text{tr} C \gamma^{(r)}$$

$$= \chi_\alpha^T (C \gamma_{m_1 \dots m_p})^T \lambda_\beta$$

$$= \chi_\alpha (\gamma_{m_1 \dots m_p}^T C^T)^{\beta\alpha} \lambda_\beta$$

$$= -\chi_\alpha (\gamma_{m_1 \dots m_p}^T C)^{\beta\alpha} \lambda_\beta$$

$$= \lambda_\beta (C \gamma)^{\beta\alpha} \chi_\alpha$$

$$= \text{tr} \bar{\lambda} \gamma_{m_1 \dots m_p} \chi$$

$$[\delta_{\xi_1}, \delta_{\xi_2}] = \dots \quad \zeta_i = -i/2$$

$$* \text{Find: } \boxed{\begin{aligned} h_{\mu\nu} &\rightarrow h_{\mu\nu} + \delta_\xi h_{\mu\nu} = h_{\mu\nu} - \frac{i}{2} \xi (\gamma_\mu \psi_\nu + \gamma_\nu \psi_\mu) \\ \psi_\mu &\rightarrow \psi_\mu + \delta_\xi \psi_\mu = \psi_\mu - i \sigma^{\rho\tau} \partial_\rho h_{\tau\mu} \xi \end{aligned}}$$

$$S^{\text{Grav}} = -\frac{i}{2} \int d^4x \varepsilon^{\mu\nu\rho\sigma} \bar{\psi}_\mu \gamma_5 \gamma_\nu \partial_\rho \psi_\sigma \\ - \frac{i}{2} \int d^4x (R_{\mu\nu}^L - \frac{1}{2} g_{\mu\nu} R^L) h^{\mu\nu}$$

Now, use noether method to

$$e^{\frac{i}{2}\xi Q}, \xi(x) : \text{Majorana}$$

$$* \mathbb{R}^4 \rightarrow \mathbb{R}$$

$$\text{and } \partial_\mu \rightarrow \tilde{\partial}_\mu$$

almost gets us there, however not quite all the way. We need to include the spin connection.

$$\delta S^{\text{grav}} = \int d^4x \bar{\psi}^\mu \partial_\mu \xi$$

$$\bar{\psi}^\mu = \frac{i}{2} \epsilon^{\mu\nu\rho\sigma} \bar{\psi}_\rho \gamma_5 \gamma_\nu \sigma^{\lambda\tau} \partial_\lambda h_{\tau\sigma}$$

add symm to overall for $\tilde{\delta}\psi_\mu$

$$\psi_\mu \rightarrow \psi_\mu + \tilde{\delta}_\xi \psi_\mu = 2\kappa^{-1} \partial_\mu \xi - i \sigma^{\rho\tau} \partial_\rho h_{\tau\mu} \xi$$

Idea: Make all terms diffeomorphism invariant

Q? Could this also be done for a conformal transformation? i.e. take a theory that is (perhaps) not and add terms which cause it to be? ...

take $d=2$:

$$S_1 = S_0 - \frac{\kappa}{4} \int d^2x \bar{\psi}^\mu \psi_\mu$$

then,

$$\rightarrow h_{\mu\nu} = e_\mu^m e_\nu^n \eta_{mn}$$

* Next order: $e_\mu^m \rightarrow e_\mu^m + \delta_\xi e_\mu^m = e_\mu^m - i \kappa \tilde{\omega}^{\rho\sigma} \gamma_{\rho\sigma} \psi_\mu$

$$\psi_\mu \rightarrow \psi_\mu + \delta_\xi \psi_\mu = \psi_\mu + 2\kappa^{-1} \tilde{D}_\mu \xi$$

$$\tilde{D}_\mu = \partial_\mu - i \tilde{\omega}_{\mu mn} \sigma^{mn} / 4$$

$$\tilde{\omega}_{\mu mn} = \omega_{\mu mn} + \frac{i\kappa^2}{4} (\bar{\psi}_m \gamma_n \psi_\mu + \bar{\psi}_\mu \gamma_m \psi_n - \bar{\psi}_\mu \gamma_n \psi_m)$$

$$\omega_{\mu mn} = \frac{1}{2} e_\mu^p (\partial_n e_{p\mu} - \partial_\mu e_{pn}) + \frac{1}{2} e_\mu^p e_n^\sigma \partial_\sigma e_{\rho\mu} e_\rho^p$$

$$\rightarrow (m \leftrightarrow n)$$

$$S = -\frac{1}{2\kappa^2} \int d^4x e R - \frac{1}{2} \int d^4x \varepsilon^{\mu\nu\rho\sigma} \bar{\Psi}_\mu \gamma_5 \gamma_\nu \tilde{D}_\rho \Psi_\sigma$$

$$\delta e_\mu^m = \kappa \bar{\xi} \gamma^m \Psi_\mu$$

$$\delta \Psi_\mu = \frac{2}{\kappa} \tilde{D}_\mu \xi$$



Now go onto Rarita-Schwinger Action:

Spin 3/2: Majorana

$$|\Psi_\mu^\alpha| = (D-1) 2^{0/2} \quad (\text{off-shell})$$

$$S = -\frac{1}{2\kappa} \int d^Dx \bar{\Psi}_\mu \gamma^{\mu\nu\rho} \partial_\nu \Psi_\rho$$

$$D=4: S = -\frac{1}{2\kappa} \int d^4x \bar{\Psi}_\mu \gamma^{\mu\nu\rho} \partial_\nu \Psi_\rho$$

$$\bullet \gamma^{\mu\nu\rho} = -i \varepsilon^{\mu\nu\rho\sigma} \gamma_\sigma \gamma_5$$

$$S = \frac{i}{2\kappa} \int d^4x \varepsilon^{\mu\nu\rho\sigma} \bar{\Psi}_\mu \gamma_\sigma \gamma_5 \partial_\nu \Psi_\rho$$

→ gauge + grav. multiplet

← $i = 1, \dots, N$

matter multiplet: $(e_\mu^a(x), \psi_\mu^i(x))$

$N=1$: supergrav. $(e_\mu^a, \psi_\mu^{\leftarrow})$ ← Majorana gravitino

✗ All exist for $D \leq 11$

→ Each dim D requires different spinors

$D=4$: $N=1, 2, \dots, 8$

$$S = S_2 + S_{3/2}$$

$$S_2 = \frac{1}{2\kappa^2} \int d^D x \, e^{\alpha\alpha} e^{\beta\beta} K_{mnab}(\omega)$$

$$S_{3/2} = -\frac{1}{2\kappa^2} \int d^D x \, e \, \bar{\Psi}_\mu \gamma^{\mu\nu\rho} \partial_\nu \Psi_\rho$$

$$D_\mu \Psi_\rho = \partial_\mu \Psi_\rho + \frac{1}{4} (\omega_\mu)_{ab} \gamma^{ab} \Psi_\rho$$

D=11 supergravity: (7) M2, M5 branes

* D=11 from dim red.

K.K. reduction on $M^D \times S^1$

h_{MN} in $M^D \times S^1$



$(h_{\mu\nu})_K$	$[20 \dots]$	$\frac{D(D-1)}{2}$	$\boxed{\square}$
$(h_{\mu D})_K$	$[10 \dots]$	D	$\boxed{\square}$
$(h_{DD})_K$	$[0 \dots 0]$	$\bullet$	$\underline{1}$

On $M^D \times S^1$ $\{ (x, y) \in M^D \times S^1 \}$

$$g_{MN} \rightarrow \left\{ \begin{array}{l} \sum_K e^{iKy/L} (h_{\mu\nu})_K(x) \\ \sum_K e^{iKy/L} (h_{\mu D})_K(x) \\ \sum_K e^{iKy/L} (h_{DD})_K(x) \end{array} \right\}$$

* More generally, we have

$$M_{D'} = M_D \times \Sigma_d$$

$$D' = D + d$$

Σ_d : compact d-dim "internal space"

- Fields in M_D theory arise from K.K. modes on Σ_d .

(i) K.K. compact: keep all massive/massless modes

(ii) Dimensional Reduction: keep only a finite set of modes

→ Usually light/massless modes.

Consistent truncation: One in which the field eqs of the omitted heavy modes are not sourced by the light modes that are kept.

T-produal Comp: $M_{D'} = M_4 \times T^{D'-4}$
 $(p_1, \dots, p_3, k_1, \dots, k_{D'-4})$

$k_i = 0$ are massless: $[21/2] = 2^5$
 $D=11$: χ : 32-comp. Majorana $2 = 2^5$

Γ^M in $11D$ | $M: D'$ $i: T$
 $\mu: 11M$

$$\Gamma^M \approx \underbrace{\gamma^m}_{4 \times 4} \otimes \underbrace{\hat{\gamma}^i}_{3 \times 3}$$

$$\Gamma^M = \gamma^m \otimes \mathbb{1}_8$$

$$\Gamma^i = \gamma_* \otimes \hat{\gamma}^i$$

$$\gamma_* = i \gamma_0 \gamma_1 \gamma_2 \gamma_3$$

$$M = 0, 1, 2, 3$$

$$i = 4, 5, \dots, 10$$

• $\psi_{m\alpha a}$, $\alpha = 1, 2, 3, 4$
 $a = 1, \dots, 8$

• M^4 : $\psi_{m\alpha a}$: transforms under Lorentz as 8 gravitinos

$$\psi_{i\alpha a}: 7 \times 8 = 56 \text{ spin } 1/2 \text{ fields}$$

(ψ_m, ψ_i) : complete fermionic content of $S_{max} = 2$ $\Rightarrow$
 $N=8$ Susy alg.

• If we started w/ a grav. in $D' \geq 12$, then dim red would contain more than eight grav.

→ Require spins $\geq 5/2$